
When Speculative Decoding Stops Paying Off at Long Context:

A Diagnostic Study with RASD

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Abstract

Speculative decoding’s $2\text{--}3\times$ short-context speedups do not reliably transfer to million-token inference. This paper asks *when and why the transfer stops*. On Llama-2-7B extended to 1 M tokens with YaRN factor 256, per-round acceptance becomes bimodal: across the measured long contexts, 48% of verify rounds fully reject the draft (pooled Wilson 95% CI $[0.39, 0.57]$) and the median per-round α is exactly 0 at 128 k and 512 k. The paired per-seed speedup against a matched target-only baseline regresses from $1.76\times [1.59, 1.93]$ at 256 k to $0.94\times [0.57, 1.30]$ at 512 k and $0.97\times [0.80, 1.14]$ at 1 M, statistical parity. A PG-19 dose-response (α : $0.71@4\text{ k} \rightarrow 0.26@8\text{ k} \rightarrow 0.11@1\text{ M}$) supports context length as the primary driver. A profiler measurement shows communication is $\leq 1.2\%$ of wall on the speculator rank, ruling out communication-overlap masking. The measured speedup is instead set by a race between two mechanisms: the acceptance collapse, and the growing cost of the speculative round itself (§5.3). Arriving at these findings required a system that runs a target-model forward at 1 M context: **RASD** combines ring-attention sequence parallelism, NF4 KV-cache quantization, YaRN scaling, and speculative verification, and reaches 1 M tokens at 39.3 GiB peak per rank on $8\times\text{A100}$, where the vanilla HuggingFace FlashAttention-2 stack OOMs at 128 k. Code, traces, and the ablation grid will be released upon acceptance.

1 Introduction

Speculative decoding [Leviathan et al., 2023, Chen et al., 2023] is the standard technique for amortizing autoregressive LLM decoding: a small draft model proposes γ tokens and the target model verifies them in one forward pass, accepting the longest matching prefix. At short context it routinely yields $2\text{--}4\times$ wall-clock speedups with provably identical output distributions, and its behavior in that regime is well understood. Its behavior at *long* context, the regime where long-context foundation models, agentic trace accumulation, and repository-scale inference actually operate, is essentially unmeasured. Prior speculative-decoding systems evaluate at short-to-moderate context [Miao et al., 2023, Cai et al., 2024, Li et al., 2024]; their acceptance rates at $10^5\text{--}10^6$ tokens are unknown.

The question this paper asks is deliberately narrow and diagnostic: *at what context length does speculative decoding stop paying off, and what mechanism makes it stop?* Answering it requires a system that can run a target-model forward at 1 M context on commodity hardware. We built such a system, **RASD** (ring attention with speculative decoding), but the system is the *vehicle* for the measurement, not the contribution. The contribution is a set of findings about the regime in which speculation is useful, three of which follow.

Finding 1 (acceptance becomes bimodal). At every long context we measure (128 k–1 M), roughly half of all verify rounds fully reject the draft ($\alpha = 0$), and most remaining rounds accept 1–2 of the 4 proposed tokens, with a small tail of fuller acceptances. The median per-round α is exactly 0 at 128 k and 512 k. The 3-seed matrix mean acceptance is 0.090 / 0.201 / 0.172 / 0.178 at 128 k / 256 k / 512 k / 1 M: non-monotone in context, so acceptance alone cannot explain the speedup shape (§5.3). The aggregate means hide a two-mode distribution, not a unimodal distribution around the mean (§5.1).

Finding 2 (context length, not prompt source, is the primary driver). A dose-response on real text (PG-19) shows mean acceptance decaying monotonically with context ($\alpha = 0.71@4\text{ k}$, $0.26@8\text{ k}$, $0.11@1\text{ M}$), so the collapse persists even when the synthetic prompt is replaced with narrative text (§5.2).

Finding 3 (communication overlap is not the mechanism). The original RASD design hypothesis was that the speculative draft pass could hide ring-attention communication latency. A torch.profiler measurement shows communication is $\leq 1.2\%$ of wall on the speculator rank at $W = 8$ over NVLink; the overlap budget does not exist at this scale. The measured speedup is set by a race between two other mechanisms: the acceptance collapse and the growing cost of the speculative round itself (§5.3).

The headline consequence. The paired per-seed speedup of RASD against a matched target-only baseline (identical infrastructure, speculative loop disabled) is $1.27 \times [0.80, 1.74]$ at 128 k, $1.76 \times [1.59, 1.93]$ at 256 k, then regresses to $0.94 \times [0.57, 1.30]$ at 512 k and $0.97 \times [0.80, 1.14]$ at 1 M (§5.3). At the headline scale, speculative decoding adds nothing; we own and quantify that rather than obscure it.

Two scope notes. First, these findings are measured on Llama-2-7B [Touvron et al., 2023] under YaRN [Peng et al., 2023] factor 256, a configuration whose base model is far out of distribution beyond its 4 k training context. That is a *deliberate* worst-case probe, not an oversight (§7): it exposes the OOD acceptance collapse in its strongest form, and it is the regime a long-context base model would encounter more gradually. Second, RASD’s capability claim (1 M-token inference at 39.3 GiB/rank) is real but secondary here; we report it in one section (§6) and otherwise treat the system as measurement infrastructure.

2 Background and Related Work

Long-context inference. Ring attention [Liu et al., 2023] shards the sequence across devices and pipelines attention through a ring of KV exchanges, making per-device memory $\mathcal{O}(N/W)$ and exact. FlashAttention [Dao et al., 2022, Dao, 2024] supplies the IO-aware local kernel; FlashDecoding [Dao et al., 2023] parallelizes single-query decode attention intra-rank. KV quantization (NF4 [Dettmers et al., 2023, 2022], KIVI [Liu et al., 2024], KVQuant [Hooper et al., 2024]) compresses the cache; vLLM [Kwon et al., 2023] manages it in paged blocks. Attention-sink prefixes [Xiao et al., 2024] and position scaling (YaRN [Peng et al., 2023], LongRoPE-style frequency search) extend short-context base models; architectural alternatives such as StripedHyena [Poli et al., 2023] and MInference [Jiang et al., 2024] avoid or sparse the attention cost entirely. RASD composes ring attention, NF4 KV quantization, YaRN, and attention sinks; none of these prior systems reports speculative-acceptance behavior at 1 M context.

Speculative decoding. Leviathan et al. [2023] and Chen et al. [2023] introduced draft-verify decoding with a provably distribution-preserving acceptance rule; expected tokens per target call is

$$\mathbb{E}[\text{tokens}] = \frac{1 - \alpha^{\gamma+1}}{1 - \alpha}, \quad (1)$$

so throughput depends on α and on the cost of the speculative round itself (§5.3). Eq. 1 is the i.i.d. idealization, in which each rejection restarts a fresh geometric trial. The standard residual-resampling acceptance rule used in implementations emits at least one token per round regardless of rejection, so the measured tokens per round is approximately $1 + \gamma\bar{\alpha}$, and Eq. 1 understates it (§5.3). Throughout, our measured α is the per-round accepted-token fraction (accepted tokens divided by drafted tokens), not the i.i.d. per-token acceptance parameter assumed by Eq. 1; the two coincide only under the equation’s independence assumptions. Medusa [Cai et al., 2024] and EAGLE [Li et al., 2024] replaced the separate draft model with heads on the target; EAGLE-2/3 improved draft

trees further. All report acceptance at short-to-moderate context. The natural mechanism question at long context is therefore: *does α itself collapse?* This paper measures it.

The overlap hypothesis. In distributed decode, each round pays a communication cost δ_{comm} for the KV rotation. Because a draft pass needs only the current KV state, it can in principle run on a separate CUDA stream while the target waits for the next block, hiding δ_{comm} entirely. This predicted an *additive* speedup beyond Eq. 1. §5.4 tests this hypothesis and rejects it at our scale, a negative result with practical value for anyone designing distributed speculative systems.

3 The RASD System

RASD (Ring Attention with Speculative Decoding) is a decode loop on $W = 8$ A100 80 GB SXM4 GPUs (NVLink-interconnected, single node). Both the target model (Llama-2-7B) and a draft model (Sheared-LLaMA-1.3B [Xia et al., 2023], which shares the target’s tokenizer) are loaded on every rank with NF4-quantized weights. The draft is deliberately not YaRN-extended: it retains its native 4k position cap and attends over only the most recent 4,096 tokens of the sequence (its KV cache covers that truncated window and is kept in bf16), while its weights are NF4-quantized like the target’s. The draft’s forward cost is therefore nearly context-independent, which matters for the mechanism analysis in §5.3. One decode round proceeds as follows: **(1)** the draft proposes $\gamma = 4$ tokens autoregressively on all ranks; **(2)** the target verifies the $\gamma + 1$ positions in a single packed forward over ring attention, with FlashAttention-2 inside each local block and KV blocks rotating via `batch_isend_irecv`; **(3)** draft and target logits are broadcast from rank 0; **(4)** every rank runs the acceptance test $\min(1, p_{\text{target}}/p_{\text{draft}})$ and resamples rejected positions from $\max(0, p - q)/Z$ [Leviathan et al., 2023], using a shared RNG seed so KV updates are bit-identical across ranks.

The memory configuration that fits 1 M context: K and V are NF4-quantized with a chunked update path (transient under 100 MiB), a bf16 attention-sink prefix of the first 128 tokens [Xiao et al., 2024] is held on rank 0 only, and position embeddings use YaRN with $f = \lceil N/4096 \rceil$ ($f = 256$ at 1 M). Per-rank peak memory is dominated by weights (≈ 4 GiB) plus the NF4 KV shard $\frac{N}{W} L 2d_h \frac{1}{2}$; measured peaks run ≈ 15 GiB above this sketch at 1 M due to transient and scratch buffers. The full memory derivation appears in the extended version (anonymized supplementary).

The design point ($\gamma = 4$, KV block size $c = 2048$, Sheared draft, prefetch depth 1) was fixed by the M3 ablation grid, five one-factor-at-a-time sweeps at 64k context with 3 seeds each. The dominant lever is block size: throughput rises $0.62 \rightarrow 0.87 \rightarrow 1.11 \rightarrow 1.25$ tok/s across $c \in \{256, 512, 1024, 2048\}$, a $2\times$ improvement, at unchanged acceptance (0.253). The other four axes fix smaller knobs; the full 49-row grid ships with the artifact release.

Baseline. Every speedup in this paper is against the *target-only* configuration: the identical RASD loop with $\gamma = 0$ (no draft model, no verify). Ring attention, NF4 KV, YaRN, and the sink prefix are identical; exactly one variable, the speculative loop, differs. This isolates the speculative mechanism from the distributed-KV infrastructure.

4 Measurement Setup

Main matrix. Four contexts $\{128, 256, 512, 1024\}$ k $\times$ three seeds $\{42, 123, 456\}$, for both RASD and target-only, with a controlled synthetic-fill protocol (a deterministic repeated-paragraph prompt that fills the context window, isolating system behavior from prompt-content variance). Each run reports throughput, mean acceptance, latency, and per-rank peak memory.

Each run targets 64 new tokens (the cap is checked before each round, so the final speculative round may overshoot by up to γ tokens; some cells log 65–66) and stops early at the first EOS token. In the degenerate YaRN regime the 128k spec cells (and one target-only seed) terminated on EOS after 16–27 tokens, versus 64–66 elsewhere. Throughput and `time_sec` include prefill (TTFT is recorded separately). Recomputing the 128k paired ratio on decode-only time, $(\text{time_sec} - \text{ttft_ms}/1000)/\text{tokens}$, gives a paired mean of $1.44\times$ [0.76, 2.12] instead of $1.27\times$; at 256k and above the convention shifts the ratio by at most $0.11\times$. The early-EOS length mismatch biases only the 128k cell; Table 3 reports the inclusive convention, matching the logged throughput.

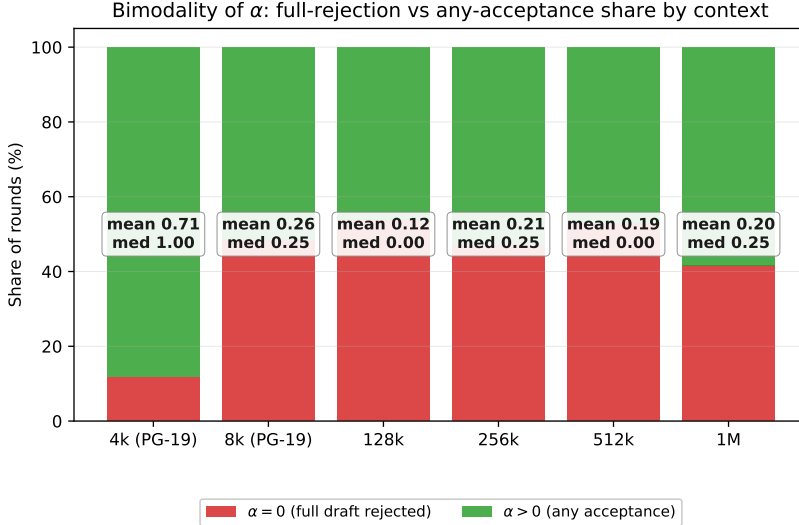


Figure 1: Per-context share of verify rounds that fully reject the draft ($\alpha = 0$) vs. accept at least one token ($\alpha > 0$), seed-42 traces. Annotations report mean and median α . At every long context roughly half of rounds fully reject the draft.

Per-round traces. For the seed-42 run at each long context we logged one JSONL sidecar per verify round (round index, position, draft tokens, accepted count), the source of the exact acceptance distributions in §5.1.

PG-19 control. Single-seed (42) runs on PG-19 narrative text [Rae et al., 2020] at 4 k and 8 k (both spec and target-only) and at 1 M (spec only), the dose-response in §5.2.

Profiler subset. torch.profiler captures at $\{32, 128, 256, 512\}$ k (single seed), bucketed into compute / comm / idle / other on rank 0. The 1 M profiler cell did not complete; see §7.

Reproducibility. All package versions are pinned in a lockfile; every number in this paper is a commit of a CSV or JSONL sidecar. Code, the lockfile, traces, and the W&B run IDs are released upon acceptance.

5 Findings

5.1 Per-round acceptance is bimodal at long context

Figure 1 and Table 1 report the per-round acceptance structure at the four long contexts, from the seed-42 traces. The aggregate means (11.5–21.3% for the seed-42 trace) are not unimodal-around-the-mean: at every context, roughly half of rounds fully reject the draft, and most remaining rounds accept 1–2 of the 4 proposed tokens, with a tail of fuller acceptances at 256 k (three rounds accept all four tokens) and 512 k (one round accepts three). The median per-round α is exactly 0 at 128 k and 512 k. Pooling all long-context rounds ($n = 119$), $P(\alpha=0) = 57/119 = 0.48$ (Wilson 95% CI $[0.39, 0.57]$). Per-context $P(\alpha=0)$, again Wilson 95%: 0.54 $[0.27, 0.81]$ at 128 k, 0.47 $[0.30, 0.64]$ at 256 k, 0.53 $[0.36, 0.69]$ at 512 k, and 0.42 $[0.26, 0.58]$ at 1 M. We report the exact discrete distribution and a binomial CI on $P(\alpha=0)$ rather than a dip test: with 13–36 rounds per context, a dip statistic would be noise.

Mechanism. In the YaRN-extrapolated regime the target’s logits are near-uniform over a large effective vocabulary (PPL = 814 at 32 k under YaRN factor 8, versus ≈ 12 at native 4 k). A 1.3B draft cannot reduce that entropy: each draft token carries probability $\sim 10^{-3}$ under the target, so verify rejects all four. On the rounds where the target does commit to a high-probability continuation, the draft tracks and 1–2 (occasionally all four) tokens are accepted. The bimodality is therefore consistent with target-logit entropy, not a defect of the verify loop: at native 4 k the same system concentrates near $\alpha = 1$ (§5.2).

Table 1: Exact counts of verify rounds by number of accepted draft tokens (of $\gamma = 4$), synthetic prompts, seed-42 traces. Columns 0/1/2/ ≥ 3 are accepted-token counts (a round accepting k tokens has $\alpha = k/4$); e.g., at 512 k the mean is $(7 \cdot 1 + 9 \cdot 2 + 1 \cdot 3)/36$ tokens per round, i.e., $\alpha = 0.194$.

Context	rounds	0	1	2	≥ 3	mean α	median α
128 k	13	7	6	0	0	0.115	0
256 k	34	16	13	2	3	0.213	0.25
512 k	36	19	7	9	1	0.194	0
1 M	36	15	13	8	0	0.201	0.25

Table 2: PG-19 dose-response (seed 42, one seed per row). Speedup is spec vs. target-only on the same prompt at the same context; no target-only 1 M PG-19 run exists. [†]No per-round trace was saved for the 1 M PG-19 run; median and $P(\alpha=0)$ are unavailable.

Context	mean α	median α	$P(\alpha=0)$	rounds	speedup
4 k	0.706	1.0	2/17	17	4.4×
8 k	0.258	0.25	15/31	31	2.3×
1 M	0.108	n/a [†]	n/a [†]	44	n/a

5.2 Context length, not prompt source, is the primary driver

A competing explanation for the collapse is the *synthetic prompt*: repeated-paragraph prompts carry no semantic momentum, so low acceptance could be a prompt artifact rather than an OOD artifact. Table 2 reports a dose-response on PG-19 narrative text. Mean α decays monotonically (0.71@4 k, 0.26@8 k, 0.11@1 M), and the bimodality appears by 8 k on real text ($P(\alpha=0) = 15/31$), even though 8 k is only 2× the model’s 4 k training window. Critically, at 1 M *real* text, acceptance falls to 0.108, at or below the synthetic-prompt 1 M value (0.201, seed 42). If the synthetic prompt were the driver, PG-19 at 1 M should recover native-context acceptance; it does not. The dose-response supports context length (base-model OOD under YaRN extrapolation) as the primary driver, with prompt source a secondary modifier. All rows are a single seed (42), an acknowledged limitation (§7).

5.3 Paired speedup regresses to parity

Table 3 reports per-seed speedup ratios $\text{RASD}_i/\text{target}_i$ for matched seeds, with t -based 95% CIs on the ratio ($df = 2$), tighter and more honest than marginal CIs on throughput. The shape is non-monotonic: a large and significant speedup at 256 k (1.76×), then a point-estimate *regression* at 512 k (0.94×) and parity at 1 M (0.97×). The 128 k and 512 k intervals straddle 1, reflecting three-seed noise on multi-hour runs; the 256 k interval excludes 1. Figure 2 plots the underlying throughputs, including the vanilla HuggingFace FlashAttention-2 ceiling (OOM at ≥ 128 k single-rank).

The honest reading: at the headline scale the speculative loop adds nothing on this base model, and the interesting question is where the payoff lives. The non-monotonic curve is a two-mechanism story, and the speedup factorizes exactly: $\text{speedup} = (\text{tokens/round, spec} \div \text{tokens/round, target-only}) \times (\text{target-only round cost} \div \text{spec round cost})$, where $\text{tokens/round} = \text{tokens_generated}/\text{n_rounds}$ and $\text{round cost} = \text{time_sec}/\text{n_rounds}$. *Mechanism (a), acceptance collapse*: α falls from 0.71 at 4 k to the 0.09–0.20 band at long contexts, capping measured tokens/round at 1.43 (128 k), 1.83 (256 k), and 1.71–1.74 (512 k–1 M), versus 3.9 at 4 k. *Mechanism (b), round-cost inflation*: the spec round costs 1.02× a target-only forward at 256 k but 1.79× at 512 k and 1.76× at 1 M. Mechanism (b), not α , produces the 256 k→512 k cliff: tokens/round barely moves (1.83 → 1.71) while the round-cost ratio nearly doubles, and flat α (0.201 / 0.172 / 0.178) coexists with a swinging speedup (1.76 / 0.94 / 0.97). Which component of the speculative round drives the inflation is not isolated: the four draft passes are nearly context-independent by construction (the draft window is capped at 4 k, §3), so the packed five-position verify and its per-round overheads are the leading candidates, hypothesis only.

Two accounting notes. First, the measured 1.76× at 256 k exceeds the Eq. 1 ceiling at $\alpha \approx 0.20$ (1.25×). The gap is the idealization gap: residual resampling emits at least one token per round,

Table 3: Per-seed paired speedup ratios $\text{RASD}_i/\text{target}_i$ at matched context and seed, the paired mean, and a t -based 95% CI ($df = 2$). The last two columns decompose the seed-42 speedup: tokens per round (spec) and the spec round cost relative to a target-only forward. Speedup = (tokens/round, spec $\div$ tokens/round, target-only) $\times$ (target-only round cost $\div$ spec round cost); the identity holds per seed. Target-only tokens/round is 1.02 in all cells except the early-EOS 128 k $s=456$ cell (1.05). Three-seed means of the round-cost ratio: 1.11 (128 k), 1.02 (256 k), 1.79 (512 k), 1.76 (1 M); tokens/round: 1.43, 1.83, 1.71, 1.74. Mean throughputs (tok/s) for reference: RASD 0.266/0.237/0.086/0.043; target-only 0.212/0.134/0.091/0.044.

Context	$s=42$	$s=123$	$s=456$	paired mean	95% CI	tokens/round	cost ratio
128 k	1.33	1.42	1.06	$1.27\times$	[0.80, 1.74]	1.54	1.14
256 k	1.82	1.78	1.69	$1.76\times$	[1.59, 1.93]	1.88	1.02
512 k	0.97	1.07	0.78	$0.94\times$	[0.57, 1.30]	1.81	1.84
1 M	1.00	1.02	0.89	$0.97\times$	[0.80, 1.14]	1.83	1.81

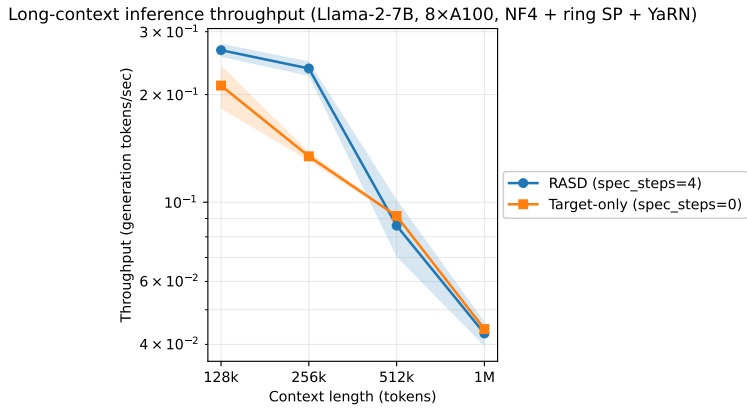


Figure 2: Throughput vs. context. RASD (blue) and matched target-only (orange) at 4 contexts $\times$ 3 seeds; vanilla HF FA-2 generate() completes only at 32 k (OOM markers at ≥ 128 k). Bands are 95% CIs (± 1.96 SEM) over seeds for display; the paired ratio CIs in Table 3 are t -based.

and acceptance is heterogeneous across rounds, so measured tokens/round (1.83) exceeds the i.i.d. prediction. Per seed, the factorization above reproduces the measured ratios exactly, leaving no residual; any residual beyond the factorization would be an infrastructure-level asymmetry that we do not isolate. Second, we report measured tokens/round throughout and treat Eq. 1 as the i.i.d. idealization only.

5.4 Communication overlap is not the mechanism

The original RASD hypothesis was that the speculative draft pass hides ring-communication latency. We test it directly. Figure 3 shows the rank-0 time-share buckets across contexts: *comm* (the sum of NCCL / c10d / batch_isend_irecv event self-times, an *issued* upper bound that includes the tick-gate barrier and therefore upper-bounds the overlap-eligible *exposed* comm) is 1.0–1.2% of wall at every context (from the 32 k canary through 512 k). Rank-0 idle is $\geq 50\%$ (synchronization covered by useful compute on the other seven ranks). The overlap budget that motivated the design does not exist at $W = 8$ on NVLink: $\delta_{\text{comm}} \ll t_{\text{verify}}$. Communication overlap is therefore not the mechanism behind any speedup in this paper; the two-mechanism decomposition of §5.3 (acceptance collapse versus round-cost inflation) is. This is a useful negative result: the ring-attention collective is already communication-efficient at this scale, and distributed speculative systems should not budget for overlap masking here. At $W \geq 32$ or over PCIe/InfiniBand the budget plausibly returns; we do not test it.

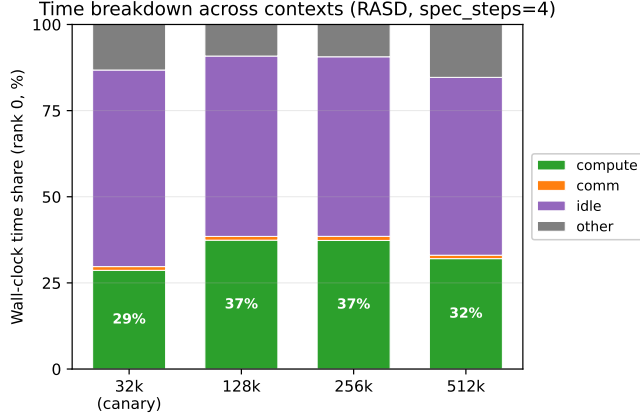


Figure 3: Rank-0 time breakdown across contexts (profiler subset, single seed). Comm is $\leq 1.2\%$ of wall at every context; idle is the synchronization share covered by compute on other ranks.

6 Capability and Design Point

One paragraph, because it is context for the findings rather than the contribution. RASD reaches 1 M-token Llama-2-7B inference at 39.3 GiB peak per rank ($8 \times A100$ 80 GB SXM4), +0.9 GiB over the target-only configuration at the same context, the cost of the draft model and its KV cache. The vanilla HuggingFace FlashAttention-2 generate() path OOMs during prefill at every context ≥ 128 k single-rank (bf16 KV cache alone is ≈ 512 GiB at 1 M), completing only the 32 k cell (30.1 GiB, 7.25 tok/s). The combination of ring-attention sharding and NF4 KV quantization is what fits the configuration; the $2 \times$ block-size lever from the M3 grid (§3) is the single most consequential engineering choice, and the remaining four sweeps fix $\gamma = 4$, the Sheared-1.3B draft, and prefetch depth 1. The full grid and the memory accounting ship with the artifact release.

7 Limitations

Parity at 1 M is owned, not hidden. The title of this paper is the finding. At 1 M the paired speedup is $0.97 \times [0.80, 1.14]$, statistical parity, and at 512 k the point estimate is a $0.94 \times$ regression whose CI straddles 1. We do not present these as wins.

Llama-2’s 4 k window is a deliberate worst-case probe. A base model trained at 4 k context and stretched by YaRN factor 256 is the strongest available OOD stress test of speculative acceptance. A long-context base model would traverse the same OOD gradient more gradually, and the dose-response (§5.2) suggests the collapse point moves out rather than disappearing. This is a hypothesis, not a demonstration; the choice of base model is the reason every finding in this paper is scoped to “Llama-2-7B under YaRN.”

Single-seed cells. The PG-19 dose-response, the profiler subset, and the per-round traces are single-seed (42). The dose-response trend is monotone and large, but the $1.76 \times$ at 256 k is the only speedup whose CI excludes 1, and three seeds is itself thin. Per-context $P(\alpha=0)$ CIs in Table 1 are wide by construction.

Small- n distributional claims. The bimodality is reported as an exact discrete distribution with a binomial CI on $P(\alpha=0)$; with 13–36 rounds per context we deliberately avoid distributional tests (e.g., dip statistics) that would not be powered.

An unrul-out mechanism. The draft and target models are independently NF4-quantized; divergence between their independently quantized logits could contribute to rejection rounds. The proposed isolation is a bf16 draft at 64 k, deferred to future work.

No production-stack baseline. We compare against a matched target-only ablation and the vanilla HF FA-2 ceiling, not vLLM / TensorRT-LLM at matched contexts; the target-only ablation is the

correct control for the speculative mechanism, but a production-stack comparison at 128 k–256 k is a follow-up.

Scope of the comm result. $\leq 1.2\%$ comm is measured on rank 0 of one 8-GPU NVLink node. Non-rank-0 profiles and multi-node configurations are unmeasured; the negative result is scoped to $W = 8$, NVLink, single node.

Missing 1 M profiler cell. The torch.profiler aggregation at 1 M did not return within 2 h of CPU work; the profiler matrix stops at 512 k, and the comm narrative extrapolates the flat 32 k–512 k trend.

8 Conclusion

We asked when speculative decoding stops paying off at long context, and measured the answer on a system built to ask the question. On Llama-2-7B under YaRN factor 256: acceptance becomes bimodal (half of rounds reject the draft entirely), the collapse is driven by context length rather than prompt source, the paired speedup regresses from $1.76\times$ at 256 k to parity at 1 M, and communication overlap, the mechanism we originally designed for, is absent at $W = 8$ on NVLink, leaving the acceptance collapse and the round-cost inflation of \$5.3 as the operative mechanisms. The practical message for long-context serving is twofold: budget speculative decoding by *measured* acceptance and round cost in the deployed context regime, not by short-context benchmarks; and on NVLink nodes, do not expect the draft pass to buy communication hiding. RASD remains as the measurement instrument: 1 M-token inference at 39.3 GiB/rank, with every trace released upon acceptance so the community can re-derive each claim.

References

- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D Lee, Deming Chen, and Tri Dao. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. *arXiv preprint arXiv:2401.10774*, 2024.
- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling. *arXiv preprint arXiv:2302.01318*, 2023.
- Tri Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations (ICLR)*, 2024.
- Tri Dao, Daniel Y Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Tri Dao, Daniel Haziza, Francisco Massa, and Grigory Sizov. Flash-decoding for long-context inference. Stanford CRFM blog, <https://crfm.stanford.edu/2023/10/12/flashdecoding.html>, 2023.
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. *arXiv preprint arXiv:2208.07339*, 2022.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. *arXiv preprint arXiv:2305.14314*, 2023.
- Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. KVQuant: Towards 10 million context length LLM inference with KV cache quantization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Huiqiang Jiang, Yucheng Li, Chengruidong Zhang, Qianhui Wu, Xufang Luo, Surin Ahn, Zhenhua Han, Amir H Abdi, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. MInference 1.0: Accelerating pre-filling for long-context LLMs via dynamic sparse attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *ACM Symposium on Operating Systems Principles (SOSP)*, 2023.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning (ICML)*, 2023.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE: Speculative sampling requires rethinking feature uncertainty. *arXiv preprint arXiv:2401.15077*, 2024.
- Hao Liu, Matei Zaharia, and Pieter Abbeel. Ring attention with blockwise transformers for near-infinite context. *arXiv preprint arXiv:2310.01889*, 2023.
- Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *International Conference on Machine Learning (ICML)*, 2024.
- Xupeng Miao, Gabriele Oliaro, Zhihao Zhang, Xinhao Cheng, Zeyu Wang, Rae Ying Yee Wong, Alan Zhu, Lijie Yang, Xiaoxiang Shi, Chunan Shi, Zhuoming Chen, Daiyaan Arfeen, Reyna Abhyankar, and Zhihao Jia. SpecInfer: Accelerating generative LLM serving with speculative inference and token tree verification. *arXiv preprint arXiv:2305.09781*, 2023.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. *arXiv preprint arXiv:2309.00071*, 2023.
- Michael Poli, Jue Wang, Eric Nguyen, Armin Thomas, Stefano Massaroli, Tri Dao, Stefano Ermon, Christopher Ré, Ce Zhang, et al. StripedHyena: Moving beyond transformers with hybrid signal processing models. Together AI technical report, <https://www.together.ai/blog/strippedhyena-7b>, 2023.
- Jack W Rae, Anna Potapenko, Siddhant M Jayakumar, Chloe Hillier, and Timothy P Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations (ICLR)*, 2020.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Mengzhou Xia, Tianyu Gao, Zhiyuan Zeng, and Danqi Chen. Sheared LLaMA: Accelerating language model pre-training via structured pruning. *arXiv preprint arXiv:2310.06694*, 2023.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *International Conference on Learning Representations (ICLR)*, 2024.